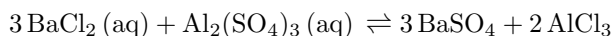


Lab 3A: Precipitation Reactions

1 Purpose

To determine the identity of the precipitate in the following double replacement reaction:



2 Procedure

2.1 Pre-lab Work

1. In a microwell plate, combine 2 drops each of BaCl_2 and $\text{Al}_2(\text{SO}_4)_3$.
2. Write a balanced chemical equation for the reaction.
3. Design a procedure that will allow you to experimentally determine the chemical formula of the precipitate, without consulting the datasheet or theory.
4. Carry out your procedure and record observations.

2.2 Lab Work

Due to a scheduling conflict, I followed Talan's procedure, which is the following:

1. Repeat step 1 from the pre-lab work, thoroughly rinsing the plate between iterations, with the following combinations:
 - $\text{BaCl}_2 + \text{Al}(\text{NO}_3)_3$
 - $\text{BaCl}_2 + \text{Na}_2\text{SO}_4$
 - $\text{FeCl}_3 + \text{Na}_2\text{SO}_4$

3 Data/Observations

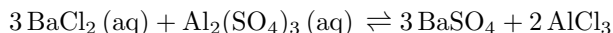
The reagents were mixed and the following observations were noted:

First Reagent	Second Reagent	Observation
BaCl_2	$\text{Al}_2(\text{SO}_4)_3$	Precipitate
BaCl_2	$\text{Al}(\text{NO}_3)_3$	No precipitate
BaCl_2	Na_2SO_4	Precipitate
FeCl_3	Na_2SO_4	No precipitate

Table 1: Observations of whether a precipitate formed.

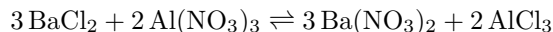
4 Analysis

Recalling the reaction equation:



The identity of the precipitate can be either BaSO_4 or AlCl_3 .

AlCl_3 can be immediately proved to not be the precipitate. In the second trial, the following reaction occurred:

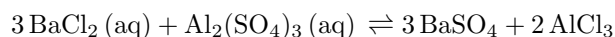


Because no precipitate formed, we know that both $\text{Ba}(\text{NO}_3)_2$ and AlCl_3 are soluble in water. Therefore, AlCl_3 cannot be the precipitate.

BaSO_4 can be proved to be the precipitate by observing the reactions and noting that a precipitate only forms when both Ba^{2+} and SO_4^{2-} are present in the reagents. In the first and third trials, both are present, and both form precipitates. In the second trial, SO_4^{2-} is absent, and no precipitate is formed. In the fourth trial, Ba^{2+} is absent, and no precipitate is formed. The absence of either leads to no precipitate forming, meaning that both ions must be present in the precipitate. Combining them, the precipitate can be deduced to be BaSO_4 .

5 Conclusion

Recalling the reaction equation:



In section 4 we concluded that the precipitate in the reaction is BaSO_4 and not AlCl_3 . When AlCl_3 was produced separately in a reaction with neither Ba^{2+} nor SO_4^{2-} , no precipitate formed. When either of the ions in BaSO_4 were absent in other trials, no precipitate formed either. Therefore AlCl_3 is not the precipitate, and BaSO_4 must be the precipitate.

5.1 Reasoning

Using our results, we can establish that AlCl_3 is soluble in water (can attain over 0.1M concentration) and that BaSO_4 has low solubility in water (precipitates before reaching 0.1M concentration).

The reason for this is because of differences in molecular attractive forces. In a soluble salt like AlCl_3 , the attractive forces between the ions and water molecules are stronger than the forces between themselves. This results in large amounts of ions being able to exist in the solution in closer proximity without dissolving.

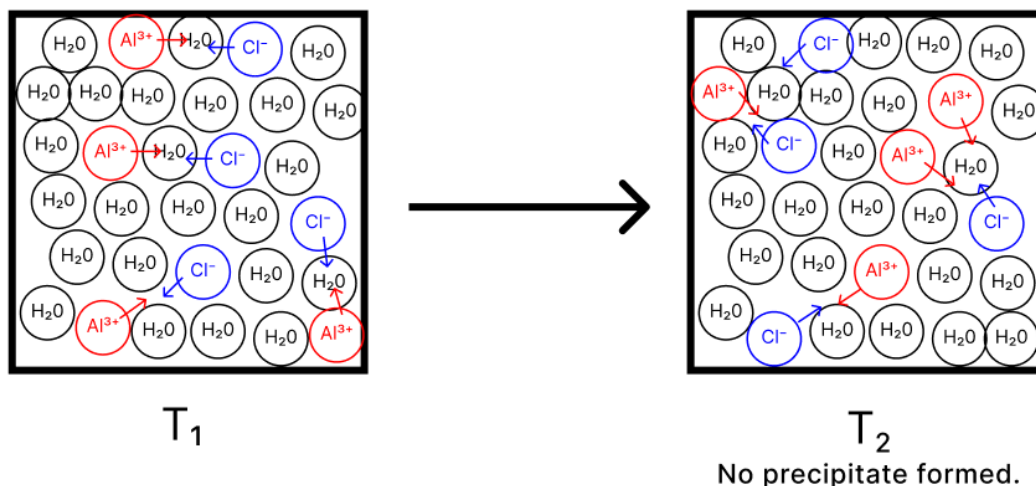


Figure 1: AlCl_3 ions in solution. The weak attractive forces between the ions allow them to exist closely together without precipitating.

In an insoluble salt like BaSO_4 , the attractive forces between the ions are stronger than the forces between the ions and water molecules. The ions are not able to exist in such close proximity as the

AlCl_3 without pulling each other out of the solution and forming a precipitate. While ions still exist in the solution, they are at a much lower concentration and are constantly being pulled in and out of the solution at equilibrium.

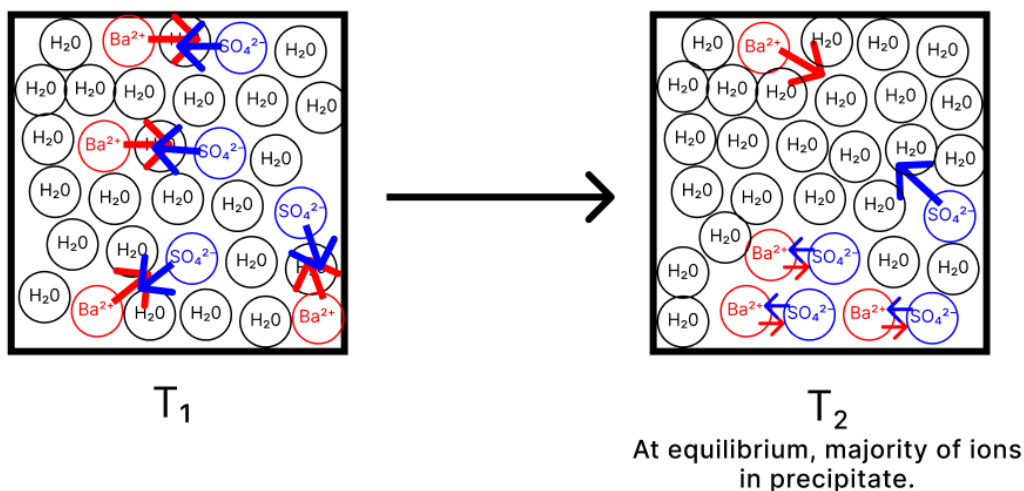


Figure 2: BaSO_4 ions in solution. The strong attractive forces between the ions cause them to precipitate out of the solution. The remaining ions must be at a much lower concentration to not precipitate. The ions are in equilibrium, and are constantly being pulled in and out of the solution.

The difference in attractive forces leads to one salt being soluble, and the other having low solubility and undergoing a chemical change to form a precipitate. By substituting ions in the reaction, we can change the attractive forces between ions, and thus change whether a precipitate forms or not. Recording the results of these changes allows us to slowly piece together the solubility table.